Program:

```
#include <stdio.h>
#define SIZE 5
int items[SIZE];
int front = -1, rear = -1;
int isFull()
{
   if ((front == rear + 1) ||   (front == 0 && rear == SIZE - 1))
      return 1;
   return 0;
}
int isEmpty()
{
   if (front == -1)
      return 1;
   return 0;
}

void enQueue(int element)
{
   if (isFull())
   {
      printf("\nQueue is full!!\n");
   }
   else
   {
      if (front == -1)  {
         front = 0;
```

```
         rear = 0;
      } else {
         rear = (rear + 1) % SIZE;
      }
      items[rear] = element;
      printf("\nInserted -> %d", element);
   }
}
int deQueue()
{
   int element;
   if (isEmpty())
   {
      printf("\nQueue is empty!!\n");
      return -1;
   }  else   {
                element = items[front];
                if (front == rear)
                {
                   front = -1;
                   rear = -1;
                }  else     {
                   front = (front + 1) % SIZE;
                }
                printf("\nDeleted element -> %d\n", element);
      return element;
   }
}
void display()
{
   int i;
   if (isEmpty())
   {
      printf("\nEmpty Queue\n");
   }
   else
   {
      printf("\nFront -> %d", front);
      printf("\nItems -> ");
      for (i = front; i != rear; i = (i + 1) % SIZE)
      {
         printf("%d ", items[i]);
      }
      printf("%d ", items[i]);
      printf("\nRear -> %d\n", rear);
   }
}
```

```
  int element;
  if (isEmpty())
  {
    printf("\nQueue is empty!!\n");
    return -1;
  }  else   {
            element = items[front];
            if (front == rear)
            {
              front = -1;
              rear = -1;
            } else    {
              front = (front + 1) % SIZE;
            }
            printf("\nDeleted element -> %d\n", element);
    return element;
  }
}
void display()
{
  int i;
  if (isEmpty())
  {
    printf("\nEmpty Queue\n");
  }
  else
  {
    printf("\nFront -> %d", front);
    printf("\nItems -> ");
    for (i = front; i != rear; i = (i + 1) % SIZE)
    {
      printf("%d ", items[i]);
    }
    printf("%d ", items[i]);
    printf("\nRear -> %d\n", rear);
  }
}
```

```
int main()
{
  deQueue();
  enQueue(1);
  enQueue(2);
  enQueue(3);
  enQueue(4);
  enQueue(5);
  enQueue(6);
  display();
  deQueue();
  display();
  enQueue(7);
  display();
  enQueue(8);
  return 0;
}
```

Output:

```
Queue is empty!!

Inserted -> 1
Inserted -> 2
Inserted -> 3
Inserted -> 4
Inserted -> 5
Queue is full!!

Front -> 0
Items -> 1 2 3 4 5
Rear -> 4
Deleted element -> 1
Front -> 1
Items -> 2 3 4 5
Rear -> 4
Inserted -> 7
Front -> 1
Items -> 2 3 4 5 7
Rear -> 0
Queue is full!!
```